The ELIXIR Cloud

Leveraging GA4GH Standards for Federated Data Analytics

Alex Kanitz (CH), Justin Clark-Casey (EMBL/EBI), Michael Crusoe (NL), Gavin Farrell (Hub), Álvaro González (FI), Thanasis Vergoulis (GR), Jonathan Tedds (Hub), and the
ELIXIR Cloud & AAI Driver Project of the Global Alliance for Genomics and Health

Audiences

- **"End users"**, e.g., researchers, healthcare professionals, patients
  - ➢ *Federated analysis of genome-scale data in highly flexible, secure cloud infrastructures*

- **Developers** and maintainers of tools, workflows and services
  - ➢ *Multidimensional integration with cloud services via (GA4GH) community standards*

- **Administrators** of compute and data centers for research
  - ➢ *On premise deployments or integration into ELIXIR or GA4GH-based hybrid and multicloud environments*

Background

GA4GH

Global Alliance for Genomics & Health

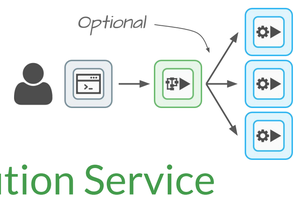
- Modular **community standards & policies** for data sharing & processing
- Brings together **all stakeholders**
  "Collaborate on standards, compete on implementations"

ELIXIR Cloud & AAI

- **GA4GH Driver Project**, led by **ELIXIR Compute Platform**
- Contribute to & implement GA4GH standards

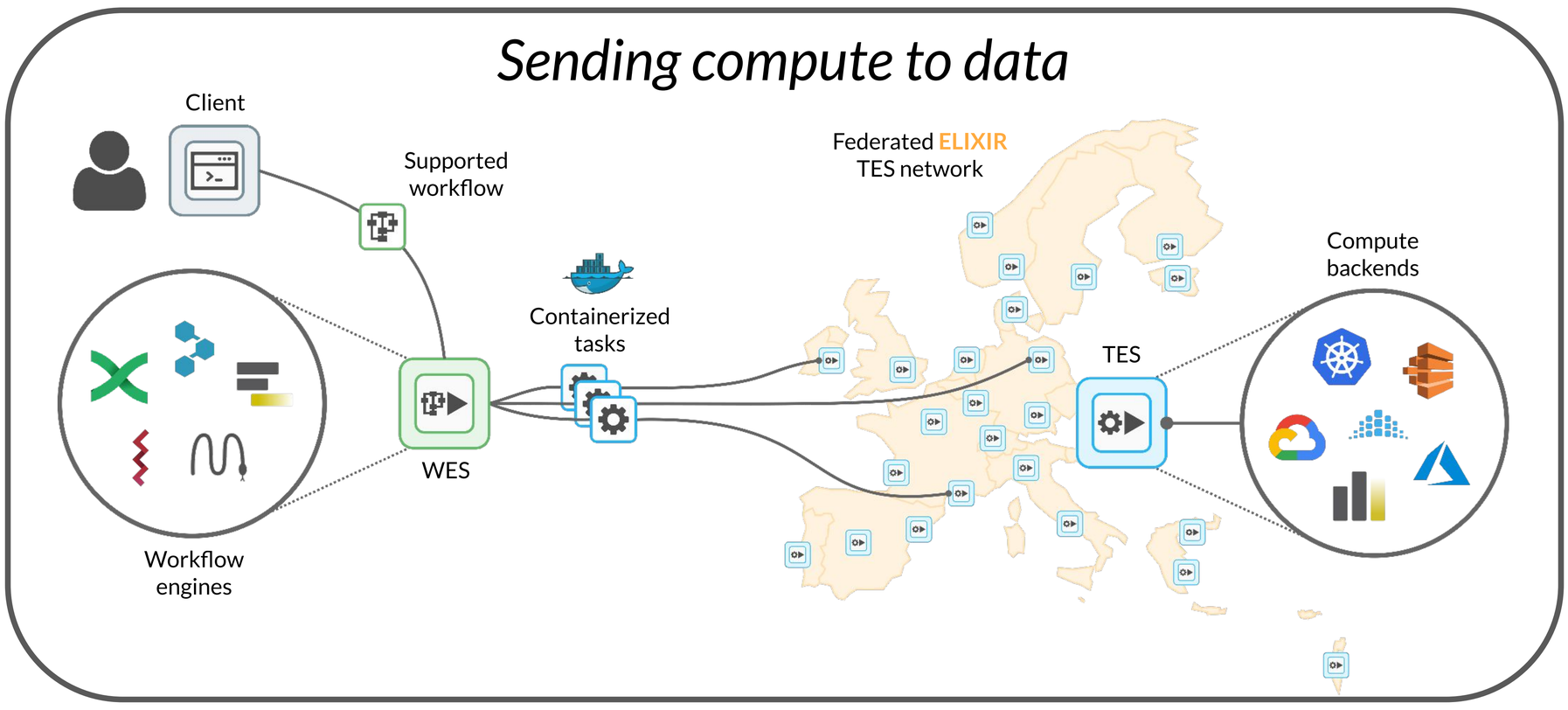
GA4GH Standards used *

* others used, omitted for clarity



- **WES** - Workflow Execution Service
  Process workflows
- **TES** - Task Execution Service
  Execute containerized tasks

Vision



Products

ELIXIR Cloud SDK

- Backend and frontend components to set up your own cloud deployment or onboard with the **ELIXIR Cloud**

ELIXIR Cloud

- A specific **ELIXIR Cloud SDK** deployment for the **ELIXIR** community

Try it out:
Run CWL, NFL & SMK
http://krini.rahtiapp.fi/

Activities

Workflow & task federation

- ➢ Use of gateways for injecting distribution middleware

ELIXIR ALL HANDS 2023 Demo @ "Reproducible Analysis & Infrastructure" mini-symposium, today 4pm

Operationalisation

- ➢ Web portal with modular, reusable client components

Initial work at: https://web-components-beta.vercel.app

Data security

- ➢ Layered encryption & isolation, operationalized by WES/TES

ELIXIR ALL HANDS 2023 Discussion @ "GA4GH Implementation Forum" workshop, tomorrow 9am

Interoperability / integration

SCHEMA, WorkflowHub, Galaxy, SAPPORO WES, Funnel, LIFE SCIENCE RI

YOU?

Contact

JOIN US ON

ELIXIR Compute Task 3 leads

Jonathan Tedds
ELIXIR Hub
jonathan.tedds@elixir-europe.org

Alex Kanitz
ELIXIR-CH
alexander.kanitz@unibas.ch

Álvaro González
ELIXIR-FI
alvaro.gonzalez@csc.fi

Thanasis Vergoulis
ELIXIR-GR
vergoulis@athenarc.gr

Michael Crusoe
ELIXIR-NL
michael.crusoe@gmail.com

Justin Clark-Casey
EMBL/EBI
justincc@ebi.ac.uk

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@ELIXIREurope

/company/elixir-europe